

Requirement Specification Document

Online Course Management System

Week 1 – Requirement Analysis and Student Database Module

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Project Title	Online Course Management System
Week	Week 1

1. Introduction

The Online Course Management System (OCMS) is a web-based application that enables educational institutions and training providers to manage online courses, student enrollments, instructors, and learning materials. The system provides a centralized database to simplify course administration and improve the online learning experience.

1.1 Purpose

The purpose of this document is to define the business, functional, and non-functional requirements of the Online Course Management System. It also specifies the database requirements for managing student information.

1.2 Project Scope

The system allows students to register, enroll in courses, access learning materials, complete assignments, and view grades. Instructors can create and manage courses, upload study materials, and evaluate student performance. Administrators oversee users, courses, and system operations.

1.3 Intended Users

- **Students** – Register, enroll in courses, access materials, submit assignments, and view results.
- **Instructors** – Create courses, upload learning resources, manage assignments, and grade students.
- **Administrators** – Manage users, courses, enrollments, and generate reports.

2. Business Requirements

- Track the status of each order (Pending, Shipped, Delivered, Cancelled)
- Record payment details for each order
- Generate reports such as total sales, top-selling products, and pending orders

3. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement Description
FR-1	The system shall allow students and instructors to register and log in securely.
FR-2	The system shall allow administrators to create, update, and delete courses.
FR-3	The system shall allow students to enroll in available courses.
FR-4	The system shall allow instructors to upload study materials and assignments.
FR-5	The system shall allow students to submit assignments online.
FR-6	The system shall allow instructors to evaluate assignments and publish grades.
FR-7	The system shall generate enrollment and academic progress reports.
FR-8	The system shall provide role-based access for students, instructors, and administrators.

4. Non-Functional Requirements

- **Performance:** The system should load pages within 3 seconds.
- **Security:** User data and course information must be securely stored.
- **Reliability:** The system should provide 99.9% availability.
- **Usability:** The interface should be simple and responsive.
- **Scalability:** The system should support thousands of users and courses.

5. Student Database Module – Data Requirements

Attribute	Data Type	Description
Student_id	INT (Primary Key)	Unique student identifier
Full_name	VARCHAR(100)	Student's full name
email	VARCHAR(100)	Student's email address
phone_number	VARCHAR(15)	Student's contact number
address	VARCHAR(100)	Name of enrolled course
registration_date	DATE	Date of enrollment

6. Assumptions and Constraints

- The system uses a relational database such as MySQL or PostgreSQL.
- Users must register before accessing the platform.

- Internet connectivity is required to use the system.
- Students can enroll only in available courses.
- This document focuses on **Week 1 – Requirement Analysis and Student Database Module**.

7. Conclusion

The **Online Course Management System** provides an efficient platform for managing online education by organizing student records, courses, enrollments, assignments, and grades. The Student Database Module forms the foundation for database design and future implementation of the complete system.